JAWAHARLAL NEHRU ENGINEERING COLLEGE

MGM UNIVERSITY, CHH. SAMBHAJINAGAR (M.S.), INDIA

MULTIMODAL SKIN CANCER DETECTION SYSTEM

USING DEEP LEARNING AND MACHINE LEARNING

B.Tech Project Report

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Submitted by

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Under the Guidance of

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CERTIFICATE

This is to certify that the project report entitled “Multimodal Skin Cancer Detection System Using Deep Learning and Machine Learning” has been prepared and submitted by Jatin Mahadik, PRN No. [Enter PRN], in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence (AI) and Data Science from the Department of Electronics and Telecommunication Engineering, Jawaharlal Nehru Engineering College, MGM University, Chh. Sambhajinagar.

The work embodied in this project report represents work carried out under the guidance and supervision of the project guide. To the best of our knowledge, the report has not been submitted to any other university or institute for the award of any degree or diploma.

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Project Approval Certificate

The practical oral examination for the above project work is conducted on _____ and the work is approved for the award of Bachelor of Technology in Artificial Intelligence (AI) and Data Science.

Internal Examiner	**External Examiner**
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I certify that the work contained in this project report, titled "Multimodal Skin Cancer Detection System Using Deep Learning and Machine Learning," is original work carried out by me under the supervision of my project guide.

I further certify that this work has not been submitted to any other institute or university for the award of any degree or diploma. Whenever information, data, theoretical analysis, figures, or text from other sources have been used, due credit has been given through citations and references.

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Jatin Mahadik

ABSTRACT

Skin cancer is a significant health concern for which timely assessment of suspicious skin lesions can support decisions about further clinical examination. Dermoscopic and clinical photographs contain visual patterns, while patient and lesion information can provide additional context. A system that considers both image-derived predictions and structured clinical features may offer a more comprehensive computer-assisted assessment than a single input source.

This project presents a multimodal skin cancer detection prototype that combines deep learning models for skin-lesion images with a machine learning model for clinical and image-derived features. EfficientNetB0 is used for binary cancer/non-cancer classification on the ISIC 2019 image dataset. A separate ResNet50 model is trained on the HAM10000 dataset. The two image models are kept as separate models because the datasets have different origins and characteristics. Their predicted cancer probabilities are then combined with structured clinical information from the PAD-UFES-20 dataset in an XGBoost fusion classifier.

The binary label mapping used for PAD-UFES-20 defines basal cell carcinoma (BCC), squamous cell carcinoma (SCC), and melanoma (MEL) as cancer, and actinic keratosis (ACK), nevus (NEV), and seborrheic keratosis (SEK) as non-cancer for this prototype. The fusion model uses 23 input features: two CNN probabilities and 21 clinical features. A defined feature schema and categorical vocabularies are used to keep inference feature order and encoding consistent with training.

The system is integrated into a web-based doctor portal through a FastAPI backend. The portal accepts a lesion image and clinical inputs, obtains predictions from the two CNNs, builds the fusion input row, and returns the XGBoost result. The backend loads model artifacts at startup and validates requests. Automated integration tests reported 16 out of 16 checks passing, including five prediction reproductions against stored test predictions with very small numerical differences. These tests validate software integration and reproducibility, not clinical safety or diagnostic performance.

The project demonstrates a modular pipeline for combining image-based deep learning with structured clinical information. The prototype is intended for academic research and decision-support experimentation only; it is not a substitute for a dermatologist, histopathology, or other standard diagnostic procedures. Future work includes external validation, broader demographic evaluation, calibration, explainability, security hardening, and prospective clinical assessment.

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NOMENCLATURE

Term	**Meaning**
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DL	Deep Learning
ML	Machine Learning
ISIC	International Skin Imaging Collaboration

HAM10000	Human Against Machine with 10000 training images dataset
PAD-UFES-20	Publicly available skin lesion dataset with clinical and image data
EfficientNetB0	Efficient convolutional neural network architecture
ResNet50	50-layer residual neural network
XGBoost	Extreme Gradient Boosting
API	Application Programming Interface
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
BCC	Basal Cell Carcinoma
SCC	Squamous Cell Carcinoma
MEL	Melanoma
ACK	Actinic Keratosis
NEV	Nevus
SEK	Seborrheic Keratosis
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

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1. INTRODUCTION

1.1 Background

Skin lesions vary in appearance and may be benign, precancerous, or malignant. Their visual characteristics can overlap, and accurate diagnosis depends on clinical expertise and, when indicated, histopathological examination. Computer vision methods can learn patterns from lesion images, while structured patient and lesion data can provide contextual signals. Deep learning has become a common approach for image classification, and gradient-boosted decision trees are effective for many tabular prediction tasks.

1.2 Problem statement

A single image classifier may overlook useful clinical context, while a tabular model cannot directly learn all visual patterns present in a lesion image. The problem addressed by this project is to develop a modular prototype that combines predictions from two independently trained image classifiers with structured clinical information and produces a final binary risk classification through a fusion model.

1.3 Need for the project

- To explore how image-based and clinical information can be combined in one prediction workflow.
- To build separate CNN models on two different public skin-lesion datasets.
- To use a tabular machine learning model to combine CNN probabilities and clinical variables.
- To integrate the trained models into a demonstrable web-based doctor portal.
- To establish reproducible preprocessing, feature ordering, input validation, and model-serving steps.

1.4 Objectives

- Prepare and use ISIC 2019 images for EfficientNetB0 binary classification.
- Prepare HAM10000 images using lesion-level splits and train a ResNet50 binary classifier.
- Prepare PAD-UFES-20 clinical data with a documented cancer/non-cancer label mapping.
- Train an XGBoost fusion model using two CNN probabilities and 21 clinical features.
- Expose inference through a FastAPI endpoint and connect it to the doctor portal.
- Verify request validation, model loading, and reproducibility against stored prediction examples.

1.5 Scope

The scope is an academic prototype for binary classification and system integration. It includes model training artifacts, feature-schema handling, backend inference, and a web interface. It does not include a clinically validated diagnostic device, automated histopathology, lesion segmentation, or a claim that model output establishes a cancer diagnosis.

1.6 Research questions

1. Can image-model probabilities and structured clinical features be combined in a single fusion classifier?
2. Can separate CNN models trained on distinct datasets be integrated consistently through a common binary output interface?
3. Can a web backend reliably validate inputs and reproduce predictions from the saved model artifacts?

1.7 Constraints and assumptions

The datasets differ in acquisition conditions, label distributions, and clinical metadata. The binary mapping simplifies several diagnostic categories and may not represent all clinical distinctions. Clinical fields must be encoded using the exact vocabularies and feature order expected by the fusion model. Missing or unfamiliar categorical values require careful handling. Performance claims must be based on held-out, appropriately designed evaluation; integration-test success alone is not evidence of clinical accuracy.

1.8 Organization of the report

Chapter 1 introduces the problem and objectives. Chapter 2 reviews relevant research. Chapter 3 describes the architecture and workflow. Chapter 4 lists the technologies used. Chapter 5 explains the conceptual design and implementation. Chapter 6 discusses the available verification results and limitations. Chapters 7 and 8 present conclusions and future work, followed by appendices and references.

2. LITERATURE SURVEY

2.1 Deep learning for skin lesion analysis

Deep convolutional neural networks learn hierarchical visual representations from images. In dermatology research, CNNs have been applied to lesion classification using dermoscopic and clinical photographs. Results depend strongly on dataset composition, image quality, label quality, split strategy, and evaluation design. Dataset leakage and differences between training and deployment populations can lead to overly optimistic results.

2.2 EfficientNet

EfficientNet uses compound scaling to balance network depth, width, and input resolution. EfficientNetB0 is a comparatively compact member of the family and is suitable when compute resources are limited. In this project, EfficientNetB0 is used as the image model for the ISIC 2019 dataset, with a binary sigmoid output. Transfer learning provides an initialization from ImageNet weights, followed by task-specific training.

2.3 ResNet50

Residual networks introduce skip connections that help optimization in deeper networks. ResNet50 is a widely used 50-layer architecture and can be initialized with ImageNet weights. This project uses ResNet50 for the HAM10000 dataset. The backbone is initially frozen while the classification head is trained, then a subset of backbone layers is fine-tuned with a smaller learning rate. The reported training setup used class weighting to place greater emphasis on the cancer class.

2.4 Clinical information and multimodal fusion

Multimodal learning combines complementary sources of information. A practical late-fusion approach first produces a prediction or representation from each modality and then combines those outputs with structured variables. Tree-based models such as XGBoost can model nonlinear relationships and interactions in tabular features without requiring all clinical fields to be transformed into dense neural embeddings. The reliability of fusion depends on consistent labels, feature construction, missing-value policy, and alignment between image predictions and the clinical record.

2.5 Related research

Reference	Contribution / relevance	Relevance to this project
[1] Esteva et al. (2017)	Demonstrated deep neural networks for skin cancer classification from clinical images.	Motivates CNN-based image classification; not a direct performance benchmark for this implementation.
[2] Tschandl et al. (2018)	Introduced HAM10000, a large dermoscopic image dataset with multiple diagnostic categories.	Source dataset for the ResNet50 branch.
[3] Codella et al. (2019)	Reported ISIC 2018 challenge tasks for lesion analysis and classification.	Provides context for standardized skin-lesion image evaluation.
[4] Pacheco et al. (2020)	Introduced PAD-UFES-20, a dataset of clinical images and patient/lesion metadata.	Source for clinical features and the fusion-model data preparation.
[5] Tan and Le (2019)	Proposed EfficientNet scaling for convolutional neural networks.	Supports selection of EfficientNetB0 as a compact image backbone.
[6] He et al. (2016)	Introduced deep residual learning for image recognition.	Supports use of ResNet50 as a residual CNN architecture.
[7] Chen and Guestrin (2016)	Introduced XGBoost, an efficient gradient-boosted tree system.	Supports the tabular fusion classifier.

2.6 Research gap

The project focuses on an engineering integration gap: building a reproducible pipeline in which two image classifiers trained on different datasets feed a clinical-data fusion model and are served through one application interface. The key challenges are consistent binary-label definitions, potential dataset shift, image/clinical pairing, feature encoding, and safe presentation of uncertainty. This is a project-

specific implementation objective rather than a claim that no previous multimodal skin-cancer research exists.

2.7 Summary

The reviewed work provides the methodological foundations for CNN-based lesion classification, transfer learning, and gradient-boosted tabular classification. This project combines these components in a modular system while recognizing that rigorous clinical validation is separate from software integration.

3. PROJECT ARCHITECTURE AND WORKFLOW

3.1 System overview

The system has three model components and a web-serving layer. EfficientNetB0 processes an image through the ISIC 2019-trained model. ResNet50 processes an image through the HAM10000-trained model. Their cancer probabilities are combined with clinical variables in the XGBoost fusion model. The backend is implemented with FastAPI, while the front end is a static doctor-portal interface.

3.2 High-level architecture

Module	Input	Processing	Output
Image branch A	Skin lesion image	Resize and EfficientNet-compatible preprocessing; EfficientNetB0 inference	Cancer probability from ISIC-trained model
Image branch B	Skin lesion image	Resize and ResNet50 Caffe-style preprocessing; ResNet50 inference	Cancer probability from HAM10000-trained model
Clinical feature builder	21 clinical fields and two CNN probabilities	Validate fields; apply training vocabularies and exact feature order	23-feature fusion row
Fusion classifier	23-feature row	XGBoost inference using configured tree range	Binary cancer/non-cancer prediction and score
API and portal	Uploaded image and form data	FastAPI validation, model loading, request handling, response formatting	Prediction response for portal display

Figure 3.1: High-level multimodal system architecture (table representation)

3.3 Dataset preparation

3.3.1 ISIC 2019 dataset

The EfficientNetB0 branch uses the ISIC 2019 JPG 224 × 224 resized dataset. After duplicate cleaning, the working split contained approximately 20,231 training images, 2,526 validation images, and 2,522 test images. The reported split counts were 3,610 cancer and 16,621 non-cancer images in training; 450 cancer and 2,076 non-cancer images in validation; and 452 cancer and 2,070 non-cancer images in testing. These counts describe the cleaned working dataset and should be rechecked against the final manifests before submission.

3.3.2 HAM10000 dataset

The HAM10000 dataset contains 10,015 images and metadata. A lesion-level split was prepared to reduce the risk of the same lesion appearing in multiple splits. With seed 42, the manifests contain 7,003 training images, 1,509 validation images, and 1,503 test images. The training split contains 1,357 cancer and 5,646 non-cancer examples; the validation split contains 294 cancer and 1,215 non-cancer examples. The test split contains 1,503 images in total; its class counts should be read from the final split manifest for any final evaluation table.

3.3.3 PAD-UFES-20 dataset

The PAD-UFES-20 metadata file contains 2,298 records and 26 columns, with all 2,298 image paths resolved during dataset inspection. The target mapping is cancer = BCC, SCC, MEL and non-cancer = ACK, NEV, SEK. The reported original diagnosis counts are BCC 845, ACK 730, NEV 244, SEK 235, SCC 192, and MEL 52. This binary mapping is a project-defined simplification for experimentation and should not be interpreted as a complete clinical taxonomy.

3.4 Data split summary

Dataset / split	Total images	Cancer	Non-cancer	Purpose
ISIC 2019 train	20,231	3,610	16,621	EfficientNetB0 training
ISIC 2019 validation	2,526	450	2,076	Model selection
ISIC 2019 test	2,522	452	2,070	Held-out evaluation
HAM10000 train	7,003	1,357	5,646	ResNet50 training
HAM10000 validation	1,509	294	1,215	Model selection
HAM10000 test	1,503	Verify from manifest	Verify from manifest	Held-out evaluation
PAD-UFES-20	2,298 records	BCC + SCC + MEL	ACK + NEV + SEK	Clinical/fusion dataset

3.5 Workflow

1. Inspect source datasets, metadata, image paths, labels, and class distributions.
2. Clean and split image data; for HAM10000, split at lesion level.
3. Train EfficientNetB0 on ISIC 2019 and save the selected model artifact.
4. Train the weighted ResNet50 model on HAM10000 and save the selected artifact.
5. Prepare PAD-UFES-20 clinical features and map diagnoses to the binary target.
6. Generate or align the two CNN probabilities with the clinical examples used for fusion training.
7. Build the 23-column fusion dataset using the exact schema and training categorical vocabularies.
8. Train and evaluate XGBoost; preserve the configuration and feature schema.
9. Load the three model artifacts in the FastAPI backend and expose the prediction endpoint.
10. Validate requests, reproduce known predictions, and review model limitations before demonstration.

3.6 Data flow and output

At inference time, the portal submits a lesion image and clinical fields to the backend. The backend validates the image and numeric/categorical inputs, obtains probabilities from both image models, creates a feature row in the exact schema order, and passes the row to XGBoost. The response is returned to the front end for display. The final output is a model-generated binary classification and score, not a confirmed diagnosis.

3.7 Important implementation controls

- Load model artifacts once at application startup rather than reloading them for every request.
- Use a fixed feature order and training-time categorical vocabularies.
- Reject missing required clinical values and invalid category values rather than silently changing their meaning.
- Validate uploaded file type and image decodability.
- Keep the model root configurable and avoid modifying the training artifacts during inference.
- Use a lock around inference where needed to avoid unsafe concurrent model access in the current CPU-oriented implementation.

4. HARDWARE, SOFTWARE AND TECHNOLOGIES USED

4.1 Hardware requirements

Training and inference were designed to run in a Python environment. The documented training setup used CPU-only execution for the HAM10000 ResNet50 script with batch size 8 to keep resource use conservative. A GPU can reduce training time, but it is not a requirement for running the CPU-oriented prototype. Exact hardware benchmark values are not included in the current project notes.

Component	**Requirement / use**
Computer	Windows development workstation
Processor	Multi-core CPU suitable for Python and TensorFlow
RAM	Sufficient memory for image batches and framework overhead; exact minimum not benchmarked
Storage	Space for datasets, split manifests, model artifacts, and logs
Optional GPU	CUDA-capable GPU for accelerated training when compatible software is installed

4.2 Software and technologies

Technology	**Role in project**
Python	Core programming language
TensorFlow / Keras	Training and inference for EfficientNetB0 and ResNet50
EfficientNetB0	ISIC 2019 image classifier
ResNet50	HAM10000 image classifier
XGBoost	Clinical and image-probability fusion classifier
Pandas / NumPy	Metadata processing and feature construction
scikit-learn	Data splitting and evaluation utilities where applicable
FastAPI	REST API for model inference
Uvicorn	ASGI server for the FastAPI application
HTML / CSS / JavaScript	Doctor-portal user interface

OpenCV / image libraries	Image decoding and preprocessing as configured by the implementation
VS Code	Development environment

4.3 Model artifact inventory

Artifact	**Dataset / purpose**	**Recorded status**
efficientnet_b0_best.keras	EfficientNetB0 trained on ISIC 2019	Artifact reported present; approximately 28.9 MB
resnet50_ham10000_weighted_best.keras	Weighted ResNet50 trained on HAM10000, class weighting 2.5:1	Artifact reported present; approximately 166.4 MB
fusion_xgboost.json	XGBoost fusion classifier	Artifact reported present; approximately 244.8 KB
feature_schema.json	Exact 23-feature order and feature definitions	Artifact reported present
training_config.json	Training-time categorical vocabularies and inference configuration	Artifact reported present

4.4 Development environment

The backend was configured to load model artifacts from a configurable root directory and to serve requests through FastAPI. The documented local server command is:

python -m uvicorn main:app --host 127.0.0.1 --port 8000

The command must be run from the backend directory with the project's virtual environment activated. The health endpoint is /health and the interactive API documentation is available at /docs when the local server is running.

4.5 Version and compatibility management

Deep-learning frameworks and their dependencies can be sensitive to Python, CUDA, cuDNN, and operating-system versions. The project environment should be preserved and documented before installing additional packages. Avoid upgrading TensorFlow, Keras, or XGBoost in the deployed environment without testing the saved artifacts and the full API flow.

5. CONCEPTUAL DESIGN AND IMPLEMENTATION

5.1 Image classification formulation

Each image model returns a sigmoid probability representing the project-defined cancer class. For a probability p and threshold t = 0.5, the binary output is cancer when p is at least t and non-cancer otherwise. The threshold is a configurable decision rule, not a clinically established cutoff. Any threshold change should be justified using validation data and the intended use case.

5.2 EfficientNetB0 branch

The EfficientNetB0 architecture is initialized with ImageNet weights and configured without its original classification top. A global average pooling layer, dropout layer, and one-unit sigmoid output form the binary classification head. The input resolution is 224 × 224 × 3. Training uses the cleaned ISIC 2019 dataset and a train/validation/test separation. The saved best model is used by the inference service.

5.3 ResNet50 branch

The ResNet50 backbone is initialized with ImageNet weights, excludes the original classification head, and receives 224 × 224 × 3 input. The project head consists of global average pooling, dropout with rate 0.5, and a sigmoid output. The training plan first freezes the backbone and trains the head, then unfreezes the final 40 backbone layers for fine-tuning at a lower learning rate. The weighted version uses a 2.5:1 cancer class weighting as recorded in the project configuration.

5.4 Clinical feature preparation

Clinical values are encoded to match the schema and training configuration. Categorical values are upper-cased and matched to the known training vocabulary. Blank or unknown categorical values may become missing values during feature construction, while the API validation layer rejects invalid categories for required inputs. Age is represented numerically. Training-time and inference-time feature construction must remain identical.

5.5 Fusion model and feature schema

The fusion classifier receives 23 features: two CNN cancer probabilities and 21 clinical features. XGBoost uses boosted decision trees to model interactions among image-model scores and clinical variables. The recorded best iteration is 74, so inference uses the first 75 trees as specified by the saved configuration.

No.	Feature	Type / notes
1	cnn_efficientnet_cancer_prob	Numeric probability from EfficientNetB0
2	cnn_resnet50_cancer_prob	Numeric probability from ResNet50

3	age	Numeric age
4	diameter_1	Numeric lesion diameter
5	diameter_2	Numeric lesion diameter
6	smoke	Categorical/binary clinical field
7	drink	Categorical/binary clinical field
8	background_father	Categorical clinical field
9	background_mother	Categorical clinical field
10	pesticide	Categorical/binary clinical field
11	gender	Categorical clinical field
12	skin_cancer_history	Categorical/binary clinical field
13	cancer_history	Categorical/binary clinical field
14	has_piped_water	Categorical/binary clinical field
15	has_sewage_system	Categorical/binary clinical field
16	fitspatrick	Categorical/ordinal field
17	region	Categorical clinical field
18	itch	Categorical/binary symptom field
19	grew	Categorical/binary symptom field
20	hurt	Categorical/binary symptom field
21	changed	Categorical/binary symptom field
22	bleed	Categorical/binary symptom field
23	elevation	Categorical/binary symptom field

5.6 Label mapping

PAD-UFES-20 diagnosis	Project binary label	Interpretation used here
BCC	1	Cancer
SCC	1	Cancer
MEL	1	Cancer
ACK	0	Non-cancer
NEV	0	Non-cancer
SEK	0	Non-cancer

This mapping follows the project definition and is intended for the binary prototype. In particular, actinic keratosis can be clinically significant and may be described as precancerous; mapping it to the non-cancer class is a task-specific simplification, not a statement that it is harmless.

5.7 Backend implementation

The FastAPI backend separates request schemas, model loading, and prediction logic. Model artifacts are loaded at startup. The /predict endpoint accepts an image and clinical fields, validates the request, executes both CNN branches, builds the fusion row, runs XGBoost, and returns a structured response. The /health endpoint reports artifact loading and feature count, while /clinical-options exposes the supported clinical option lists for the form.

5.8 Front-end integration

The static portal contains a prediction form and a results page. The form submits the image and clinical fields to the backend. The result page reads the live response and presents the returned prediction. Any patient lists or authentication data implemented with browser localStorage are demo-only and must not be represented as secure clinical records or real authentication.

5.9 Validation and error handling

Condition	Expected handling
Required clinical field missing	Return validation error (422)

Invalid categorical value	Return validation error (422)
Invalid numeric value	Return validation error (422)
Missing image	Return request error (400)
Non-image upload	Reject request (400)
Corrupt or truncated image	Reject request (400)
Model artifact unavailable	Health check should identify service/model loading problem

5.10 Privacy and safety considerations

Clinical images and patient details are sensitive. A production deployment would require authentication, authorization, secure transport, access logging, data retention rules, encryption, and a threat model. The current local prototype is for academic demonstration. Predictions must be communicated as decision-support outputs with uncertainty and a clear instruction that clinical evaluation remains necessary.

6. RESULTS AND DISCUSSION

6.1 Dataset preparation results

The ISIC 2019 cleaned working split contains 25,279 images in total across training, validation, and test sets. The HAM10000 lesion-level manifests contain 10,015 images across three splits. PAD-UFES-20 contains 2,298 metadata records with resolved image paths. These totals provide a basis for reproducible experiments, but final reports should confirm counts directly from the final manifest files.

6.2 Model and integration verification

The available project verification notes report a successful end-to-end backend test run with 16 out of 16 checks passing. The health endpoint confirmed that all four required model/configuration artifacts loaded and that the fusion model used 23 features. Five stored test examples were reproduced with maximum absolute differences reported as 1.8×10^{-9} for the fusion output, 5.6×10^{-7} for EfficientNetB0, and 1.4×10^{-6} for ResNet50.

Verification item	**Observed result**	**Meaning**
Backend end-to-end checks	16/16 PASS	The tested integration paths completed successfully.

Artifact health check	All 4 artifacts loaded	Model/configuration files were found and loaded in the tested environment.
Fusion feature count	23	Input row matched the configured two-model plus clinical schema.
Fusion reproduction (5 examples)	Maximum difference 1.8×10^{-9}	Near-identical stored output for the reproduced examples.
EfficientNet reproduction (5 examples)	Maximum difference 5.6×10^{-7}	Near-identical stored output for the reproduced examples.
ResNet50 reproduction (5 examples)	Maximum difference 1.4×10^{-6}	Near-identical stored output for the reproduced examples.
Missing/invalid inputs	Expected 400/422 responses	Request validation rejected the tested malformed inputs.

6.3 Interpretation

These results support that the tested API and inference pipeline were connected correctly and that selected predictions could be reproduced from saved artifacts. They do not provide sensitivity, specificity, precision, recall, F1-score, ROC-AUC, calibration, confidence intervals, or clinical validation results for the full held-out datasets. Those metrics must be computed from the final test manifests and ground-truth labels before making claims about classification performance.

6.4 Evaluation metrics recommended

Metric	**Definition / purpose**
Accuracy	(TP + TN) / (TP + TN + FP + FN); overall correctness, but can hide minority-class errors.
Precision	TP / (TP + FP); proportion of predicted cancer cases that are positive.
Recall / Sensitivity	TP / (TP + FN); proportion of actual cancer cases detected.
Specificity	TN / (TN + FP); proportion of non-cancer cases correctly identified.
F1-score	Harmonic mean of precision and recall.
ROC-AUC	Ranking discrimination across thresholds.

Confusion matrix	Counts of TP, TN, FP, and FN.
Calibration	Agreement between predicted probabilities and observed outcome frequencies.

6.5 Limitations

- The image models are trained on different datasets, and their probability outputs may not be calibrated to the same population.
- The fusion dataset's image probabilities must be generated in a leakage-aware manner; training predictions should ideally be out-of-fold or generated from models that did not train on those examples.
- Dataset label mapping compresses multiple diagnoses into two classes and treats ACK as non-cancer by project definition.
- Dataset shift, skin tone representation, image acquisition differences, and demographic imbalance may affect performance.
- Integration test reproduction is not equivalent to a robust clinical evaluation.
- The current portal's localStorage demo authentication and patient records are not suitable for production use.

6.6 Discussion

The main engineering outcome is a working modular pipeline with a documented feature schema and request validation. The use of two independent CNN branches allows experimentation with different image datasets, while XGBoost provides a clear interface for combining their outputs with clinical variables. The next evaluation step should prioritize a leakage audit, held-out metrics for each image model and the fusion model, subgroup analysis, and probability calibration.

7. CONCLUSIONS

This project developed a multimodal skin cancer detection prototype that combines two image classifiers with clinical variables through an XGBoost fusion model. EfficientNetB0 was assigned to the ISIC 2019 dataset, ResNet50 to HAM10000, and PAD-UFES-20 to clinical feature preparation and fusion. The final schema contains 23 features and is shared by the training and inference workflows.

The models were integrated into a FastAPI backend and connected to a web-based doctor portal. The available integration checks reported 16 out of 16 passing, including model-artifact loading, input validation, and reproduction of selected stored predictions with very small numerical differences. This establishes that the tested software components operate together as intended in the tested environment.

The current evidence is limited to implementation and reproducibility checks. Full held-out classification metrics and independent clinical validation are required before the model can be considered reliable for any medical use. The prototype should be used only for academic demonstration and not as a replacement for professional assessment.

8. FUTURE SCOPE

1. Compute complete held-out metrics for both CNN models and the fusion model, including sensitivity, specificity, precision, recall, F1-score, ROC-AUC, and confusion matrices.
2. Use out-of-fold CNN predictions or a strictly separated fusion-training protocol to prevent leakage when building the fusion dataset.
3. Perform external validation on datasets from different institutions and imaging conditions.
4. Assess performance across skin tones, age groups, anatomical regions, and other relevant subgroups.
5. Calibrate probability outputs and select thresholds based on intended use, validation evidence, and the relative costs of false negatives and false positives.
6. Add explainability methods, such as image saliency or feature attribution, with careful assessment of their reliability.
7. Improve user-facing uncertainty communication and ensure the portal never presents model output as a definitive diagnosis.
8. Replace demo authentication and localStorage patient records with secure identity management, role-based access, audit trails, and compliant data handling.
9. Add versioned model registry, structured logging, monitoring, and rollback support for future model updates.
10. Conduct prospective clinical studies and obtain appropriate institutional, ethical, and regulatory review before real-world deployment.

9. APPENDICES

Appendix A: API endpoint summary

Endpoint	**Purpose**	**Notes**
GET /health	Check backend and artifact loading	Use before starting a demonstration.
GET /clinical-options	Return supported categorical form options	Options should match training vocabularies.
POST /predict	Accept image and clinical fields and return prediction	Requires valid image and required clinical inputs.
GET /docs	Interactive API documentation	Provided by FastAPI when server is running.

Appendix B: Local server startup

1. Open PowerShell or the VS Code terminal.

2. Change directory to the project's backend folder.
3. Activate the project's virtual environment or call its Python executable directly.
4. Start Uvicorn with: python -m uvicorn main:app --host 127.0.0.1 --port 8000
5. Open http://127.0.0.1:8000/health to verify that the backend is ready.
6. Open http://127.0.0.1:8000/docs to inspect the API.
7. Open the static portal page through the project's configured front-end method and submit a test image with valid clinical inputs.

Appendix C: Known limitations and mitigations

Limitation	Recommended mitigation
Different source datasets	Document source, preprocessing, and population differences; validate externally.
Potential fusion leakage	Generate out-of-fold or strictly held-out CNN predictions for fusion training.
Binary label simplification	State the label mapping clearly and evaluate alternatives with domain experts.
Uncalibrated probabilities	Use calibration on a validation set and report calibration metrics.
Demo authentication	Do not use for real patient data; implement secure authentication and access controls.
No clinical validation	Restrict to research/demo use until appropriate validation and approvals are complete.

Appendix D: Items to complete before final submission

- Replace [Enter PRN] with the correct university PRN.
- Replace [Project Guide Name] with the guide's official name and designation.
- Confirm the exact department name and institution wording required by the department.
- Insert the official MGM University logo if required by the approved format.
- Add actual training curves, confusion matrices, ROC curves, and final held-out test metrics after computing them from the final manifests.
- Verify the test class counts from the HAM10000 test manifest.
- Update table of contents and page numbers after all figures and appendices are finalized.
- Confirm that every citation used in the report appears in the references and that the reference numbering follows first appearance in the report.

10. REFERENCES

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